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United States Patent	12411116
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Furuya; Takaaki et al.

Apparatus with gas detection function

Abstract

Provided is an apparatus with a gas detection function capable of detecting a detection target gas with high accuracy. An apparatus with a gas detection function comprises: a housing (10); a vibration source (20); and a gas measurement unit (40) located inside the housing and demarcated by a partition, wherein the vibration source is located outside the gas measurement unit, the gas measurement unit includes a detector (41) located on a substrate (30), and a gas detection space (42) provided with a hole (43) through which a gas passes, and a frequency f expressed by the following Formula (1): $f = \frac{c}{2L} \sqrt{\frac{S}{V}}$ Formula(1) is 500 Hz or more, where V is a volume of the gas detection space, S is a cross-sectional area of the hole, L is an effective length of the hole, and c is a sound speed.

Inventors:	Furuya; Takaaki (Tokyo, JP), Ugomori; Naoya (Tokyo, JP)
Applicant:	Asahi Kasei Microdevices Corporation (Tokyo, JP)
Family ID:	87431844
Assignee:	Asahi Kasei Microdevices Corporation (Tokyo, JP)
Appl. No.:	18/163282
Filed:	February 02, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230243796 A1	Aug. 03, 2023

Foreign Application Priority Data

JP	2022-015244	Feb. 02, 2022
JP	2023-009604	Jan. 25, 2023

Publication Classification

Int. Cl.: G01N33/00 (20060101); G01N29/036 (20060101); G01N29/24 (20060101)

U.S. Cl.:

CPC G01N33/005 (20130101); G01N29/036 (20130101); G01N29/2431 (20130101); G01N2291/021 (20130101); G01N2291/02809 (20130101)

Field of Classification Search

CPC: G01N (33/005); G01N (29/036); G01N (29/2431); G01N (2291/021); G01N (2291/02809); G01N (29/222)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6378372	12/2001	Karr	73/24.01	G01N 29/36
2003/0037801	12/2002	Bailey	134/1	H01L 21/6875
2010/0263437	12/2009	Koyama	73/64.53	G01N 29/2443
2015/0097663	12/2014	Sloo et al.	N/A	N/A
2015/0338390	12/2014	Anglin, Jr.	73/23.3	G01N 27/227
2017/0176330	12/2016	Lahti	N/A	G01N 33/497

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
S5334581	12/1977	JP	N/A
S5954950	12/1983	JP	N/A
H06510870	12/1993	JP	N/A
H0783756	12/1994	JP	N/A
2006138752	12/2005	JP	N/A
2019091489	12/2018	JP	N/A
9215088	12/1991	WO	N/A
WO-2016109781	12/2015	WO	G01N 33/0036

Primary Examiner: Gray; Francis C

Attorney, Agent or Firm: KENJA IP LAW PC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to and the benefit of Japanese Patent Application No. 2022-015244 filed Feb. 2, 2022 and Japanese Patent Application No. 2023-009604 filed Jan. 25, 2023, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to an apparatus with a gas detection function.

BACKGROUND

(3) Gas sensors that detect gas are used in various fields. With the progress of semiconductor technology and MEMS technology, gas sensors are increasingly miniaturized, and mounting of gas sensors in housings of apparatuses that emit sound, such as smart home devices, is promoted (see, for example, PTL 1).

CITATION LIST

Patent Literature

(4) PTL 1: JP 2019-91489 A

SUMMARY

(5) In the apparatus disclosed in PTL 1, the gas sensor is exposed to air vibration such as music and voice coming from the speaker in the same housing. The gas sensor is also exposed to air vibration from the motor of the air conditioner or the like. These may cause the resonance of the gas in the gas sensor and excite vibration. Noise caused by such vibration degrades the detection accuracy of the gas sensor. In the case where the light source, detector, etc. used in the gas sensor include microstructures suspended in the air using MEMS technology, the influence of noise caused by vibration is particularly significant, and the detection accuracy is likely to degrade greatly.

(6) It could therefore be helpful to provide an apparatus with a gas detection function capable of detecting a detection target gas (i.e. a gas to be detected) with high accuracy.

(7) [1] An apparatus with a gas detection function according to an embodiment of the resent disclosure comprises: a housing; a vibration source; and a gas measurement unit located inside the housing and demarcated by a partition, wherein the vibration source is located outside the gas measurement unit, the gas measurement unit includes a detector located on a substrate, and a gas detection space provided with a hole through which a gas passes, and a frequency f expressed by the following Formula (1):

$$(8) \quad f = \frac{c}{2} \sqrt{\frac{S}{VL}} \quad \text{Formula(1)}$$

(9) is 500 Hz or more, where V is a volume of the gas detection space, S is a cross-sectional area of the hole, L is an effective length of the hole, and c is a sound speed.

(10) [2] As an embodiment of the resent disclosure, in [1], the hole is provided with a vent pipe, and the effective length is given by the following Formula (2):

$$L = L' + 1.5a \quad \text{Formula (2),}$$

(11) where L' is a total length of the hole and the vent pipe, and a is a radius of the hole.

(12) [3] As an embodiment of the resent disclosure, in [1], the hole is provided with a vent pipe, an exit of the hole and the vent pipe is flat, and the effective length is given by the following Formula (3):

$$L = L' + 1.7a \quad \text{Formula (3),}$$

(13) where L' is a total length of the hole and the vent pipe, and a is a radius of the hole.

(14) [4] As an embodiment of the resent disclosure, in [1], the hole is provided with a vent pipe, a cross-sectional shape of the hole and the vent pipe is irregular, and the effective length is given by the following Formula (4):

$$L = L' + \sqrt{\text{square root over } (S_{\text{sub.out}})} \quad \text{Formula (4),}$$

(15) where L' is a total length of the hole and the vent pipe, and $S_{\text{sub.out}}$ is an area of an exit opening of the hole and the vent pipe.

(16) [5] As an embodiment of the resent disclosure, in any of [1] to [4], the frequency f is 1.0 kHz or more.

(17) [6] As an embodiment of the resent disclosure, in any of [1] to [5], the frequency f is 2.0 kHz or more.

(18) [7] An apparatus with a gas detection function according to an embodiment of the resent

disclosure comprises: a housing; a vibration source; and a gas measurement unit located inside the housing, wherein the gas measurement unit includes a detector located on a substrate, and a gas detection space provided with a hole through which a gas passes, the hole is provided with a vent pipe, a cross-sectional shape of the hole and the vent pipe is irregular, an effective length L of the hole is given by the following Formula (5):

$$L=L'+\sqrt{S_{\text{sub.out}}} \quad \text{Formula (5),}$$

(19) where L' is a total length of the hole and the vent pipe, and S.sub.out is an area of an exit opening of the hole and the vent pipe, and a frequency f expressed by the following Formula (6):

$$(20) \quad f = \frac{c}{2} \sqrt{\frac{S}{VL}} \quad \text{Formula(6)}$$

(21) is 100 Hz or less, where V is a volume of the gas detection space, S is a cross-sectional area of the hole, and c is a sound speed.

(22) [8] As an embodiment of the resent disclosure, in any of [1] to [7], the vibration source or the detector is located at a plane of symmetry of a rectangular parallelepiped approximately defined for the housing.

(23) [9] As an embodiment of the resent disclosure, in any of [1] to [8], the vibration source or the detector is located at an axis of symmetry of a cylinder approximately defined for the housing.

(24) [10] As an embodiment of the resent disclosure, in any of [1] to [9], the vibration source or the detector is located at a center point of a sphere approximately defined for the housing.

(25) [11] As an embodiment of the resent disclosure, in any of [1] to [10], the vibration source or the detector is located at an end of a rectangle approximately defined for the substrate.

(26) [12] As an embodiment of the resent disclosure, in any of [1] to [11], a frequency f.sub.hus expressed by the following Formula (7):

$$(27) \quad f_{\text{hus}} = \frac{c}{L_{\text{hus}}} \quad \text{Formula(7)}$$

(28) is 500 Hz or more, where L.sub.hus is a typical length of the housing.

(29) [13] As an embodiment of the resent disclosure, in [12], the frequency f.sub.hus is 1.0 kHz or more.

(30) [14] As an embodiment of the resent disclosure, in [12] or [13], the frequency f.sub.hus is 2.0 kHz or more.

(31) [15] As an embodiment of the resent disclosure, in any of [1] to [14], a frequency f.sub.sub expressed by the following Formula (8):

$$(32) \quad f_{\text{sub}} = \frac{h}{2L_{\text{sub}}^2} \sqrt{\frac{E_{\text{g}}}{3(1-\sigma^2)}} \quad \text{Formula(8)}$$

(33) is 500 Hz or more, where S.sub.sub is an area of the substrate, E is a modulus of longitudinal elasticity of the substrate, h is a thickness of the substrate, σ is Poisson's ratio, γ is a dimensionless constant of 3.65, and a typical length L.sub.sub is a square root of the area of the substrate.

(34) [16] As an embodiment of the resent disclosure, in [15], the frequency f.sub.sub is 1.0 kHz or more.

(35) [17] As an embodiment of the resent disclosure, in [15] or [16], the frequency f.sub.sub is 2.0 kHz or more.

(36) [18] As an embodiment of the resent disclosure, in any of [1] to [17], the hole is provided with a dust filter, and an acoustic characteristic impedance of the dust filter is 4000 kg/m.sup.2s or more.

(37) [19] As an embodiment of the resent disclosure, in any of [1] to [18], the detector includes an air-suspended microstructure.

(38) [20] As an embodiment of the resent disclosure, in any of [1] to [19], the vibration source is located inside the housing.

(39) It is thus possible to provide an apparatus with a gas detection function capable of detecting a detection target gas with high accuracy.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the accompanying drawings:

- (2) FIG. 1 is a diagram illustrating an example of the structure of an apparatus with a gas detection function according to an embodiment;
- (3) FIG. 2 is a diagram illustrating an example of the structure of an apparatus with a gas detection function in which a hole is provided with a vent pipe;
- (4) FIG. 3 is a diagram illustrating an example of the structure of an apparatus with a gas detection function in which a hole is provided with a dust filter;
- (5) FIG. 4 is a diagram illustrating an example of noise intensity numerical calculation results; and
- (6) FIG. 5 is a diagram illustrating another example of noise intensity numerical calculation results.

DETAILED DESCRIPTION

(7) An apparatus with a gas detection function according to an embodiment of the present disclosure will be described below, with reference to the drawings. In the drawings, the same or corresponding parts are given the same reference signs. In the following description of embodiments, the description of the same or corresponding parts is omitted or simplified as appropriate.

(8) <Apparatus>

(9) FIG. 1 is a diagram illustrating the structure of an apparatus 1 with a gas detection function according to this embodiment. The gas detection function is, for example, a function of detecting the concentration in the air of a detection target gas to be detected. The apparatus with the gas detection function includes a housing 10, a vibration source 20, a substrate 30, and a gas measurement unit 40. The gas measurement unit 40 includes a detector 41 and a gas detection space 42. The gas detection space 42 is provided with a hole 43 through which gas passes. The gas mentioned herein is, for example, air, and may include the detection target gas. In the following description, the gas is assumed to be air. The apparatus 1 may include a circuit 50 as in this embodiment. The apparatus 1 may further include a display unit such as a display.

(10) In the apparatus 1 according to this embodiment, the vibration source 20 is attached to the housing 10, and the gas measurement unit 40 is located inside the housing 10. The gas measurement unit 40 is demarcated by a partition in the housing 10, and the gas detection space 42 is located inside the demarcated gas measurement unit 40. In the apparatus 1, the gas measurement unit 40 and the circuit 50 are arranged on the substrate 30. The gas measurement unit 40 includes the detector 41 located on the substrate 30, and the gas detection space 42 provided with the hole 43. The gas detection space 42 may be the internal space of the partition in part of which the hole 43 is formed.

(11) The apparatus 1 is electronic equipment in particular, and includes any of various apparatuses such as electrical products. The apparatus 1 may be electronic equipment having an acoustic function of performing communication, amusement, information notification, and the like by sound such as voice and music. Specific examples of such apparatus 1 include a music player, a smart speaker, a smart home device, a smartphone, and an earphone. The apparatus 1 may be an apparatus that provides an air conditioning function. Specific examples of such apparatus 1 include an air conditioner, an air purifier, a ventilation fan, and a heating, ventilation, and air conditioning (HVAC) apparatus. The acoustic function or the air conditioning function is independently performed according to, for example, user operation, regardless of whether the gas detection function is operating. There is thus a possibility that, in the detection of the detection target gas by the gas measurement unit 40, noise increases due to the influence of irregular sound, vibration, etc. (hereinafter referred to as “unwanted sound”).

(12) <Housing>

(13) The housing **10** is the exterior of the apparatus **1**, and has a function of holding the vibration source **20**. The housing **10** may also hold the substrate **30**. The housing **10** may be made of metal, glass, resin, or a composite material thereof. The housing **10** may include a switch, a power connector, a communication connector, a display unit, an antenna, an input interface, a camera, a light detection and ranging (LiDAR) sensor, an infrared camera, a microphone, an indicator lamp, a fin, and the like.

(14) <Vibration Source>

(15) The vibration source **20** generates vibration as the acoustic function or the air conditioning function is performed. In this embodiment, the vibration source **20** emits sound. A specific example of such vibration source **20** is a speaker. Alternatively, the vibration source **20** may circulate air. A specific example of such vibration source **20** is a motor. The vibration source **20** is located outside the gas measurement unit **40**. Although the vibration source **20** is attached to the housing **10** in this embodiment, the attachment position is not limited to such. For example, the vibration source **20** may be attached to the substrate **30**. The vibration source **20** may be located inside or outside the housing **10**.

(16) <Substrate>

(17) The substrate **30** has a function of holding the gas measurement unit **40**. In this embodiment, the substrate **30** also holds the circuit **50**. The substrate **30** may also hold the vibration source **20**. The material of the substrate **30** is, for example, paper, glass cloth, polyimide film, PET film, ceramics, or the like. Examples of the resin include phenol resin, epoxy resin, polyimide resin, bismaleimide triazine resin, fluorine resin, and polyphenylene oxide resin.

(18) <Gas Measurement Unit>

(19) The gas measurement unit **40** detects the detection target gas. In detail, the gas measurement unit **40** measures the concentration of the detection target gas in the air, and outputs an electrical signal indicating the measurement result. The gas measurement unit **40** may be a non-dispersive infrared absorption type gas concentration measurement device having a known structure. Examples of the detection target gas include carbon dioxide, water vapor, carbon monoxide, nitrogen monoxide, ammonia, sulfur dioxide, alcohol, formaldehyde, and hydrocarbon-based gases such as methane and propane. The gas measurement unit **40** includes the detector **41**, the gas detection space **42**, and the hole **43**.

(20) <Detector>

(21) The detector **41** detects changes depending on the amount of the detection target gas present, in the measurement of the concentration of the detection target gas contained in the gas (air) in the gas detection space **42**. In this embodiment, the detector **41** includes a light emitting element and a light receiving element. The light emitted from the light emitting element passes through the gas detection space **42**, is absorbed and attenuated depending on the amount of the detection target gas present, and is received by the light receiving element. The amount of the detection target gas present can be measured from the attenuation of the light. Examples of the light emitting element include a light emitting diode (LED), a lamp, a laser (light amplification by stimulated emission of radiation), an organic light emitting element, a microelectromechanical systems (MEMS) heater, and a vertical-cavity surface-emitting laser (VCSEL). Examples of the light receiving element include a photodiode, a phototransistor, a thermopile, a pyroelectric sensor, and a bolometer. The detector **41** may include an air-suspended microstructure using MEMS technology. The detector **41** may include a resin material, a metal package, or the like as a protective member that protects the light emitting element and the light receiving element. The detector **41** may include an optical filter that limits the wavelength of light.

(22) As another example, the detector **41** may include a heating element and a resistor. For example, the heat generated from the heating element by applying electric current is transferred to the detection target gas in the gas detection space **42**, and the temperature increase of the heating element depending on the amount of the detection target gas present is detected as a change in

resistance. Examples of the heating element include a MEMS heater and a lamp. As another example, the detector **41** may include a light emitting element and a pressure sensitive element. The light emitted from the light emitting element is absorbed depending on the amount of the detection target gas present in the gas detection space **42**, and the temperature, pressure, etc. of the air in the gas detection space **42** increase. The amount of the detection target gas present can be measured from the pressure increase detected by the pressure sensitive element. Examples of the pressure sensitive element include a MEMS pressure sensor and a microphone.

(23) <Gas Detection Space>

(24) The gas detection space **42** is separated by an outer wall, and has a function of containing gas such as air in the internal space. The gas contained in the gas detection space **42** is replaced through the hole **43**. The outer wall of the gas detection space **42** is made of metal, resin, or the like.

(25) <Hole>

(26) The hole **43** is a hole formed in the outer wall of the gas detection space **42**. Gas passes through the hole **43** and replaces the gas in the gas detection space **42**. The number of holes **43** may be more than one.

(27) As illustrated in FIG. 2, the hole **43** may be provided with a vent pipe **44**. The vent pipe **44** has a function of limiting the space through which gas passes and guiding the gas to the gas detection space **42**. The vent pipe **44** may be an additional part, may be a part of the housing **10** connected to the external space in a tubular shape, or may be formed by extending the hole **43** in a tubular shape. As illustrated in FIG. 3, the hole **43** may be provided with a dust filter **45** for dust prevention.

(28) <Circuit>

(29) The circuit **50** may control the overall apparatus **1**. For example, the circuit **50** may control the vibration source **20**. For example, the circuit **50** may perform computation on the output signal from the gas measurement unit **40** in order to obtain the concentration of the detection target gas. The circuit **50** may include at least one processor. The processor is, for example, a general-purpose processor or a dedicated processor specialized for specific processing. The processor is, however, not limited to such, and may be any processor.

(30) <Explanation of Principle>

(31) Gas is replaced, that is, taken in and out of the gas detection space **42**, through the hole **43**. Resonance occurs as a result of gas being taken in and out at around a specific frequency. This is the same as the Helmholtz resonance phenomenon in which a hollow musical instrument with holes, such as a guitar or an ocarina, or a bottle resonates in specific scale due to external vibration and airflow. The Helmholtz resonance phenomenon is a phenomenon in which the gas inside the cavity behaves as an air spring and the gas around the hole, as a mass, behaves as an oscillator having a resonance frequency. The gas present in the gas detection space **42** vibrates due to the Helmholtz resonance phenomenon. Coarseness and fineness caused by such vibration are detected by the detector **41** as variation in the concentration of the detection target gas, causing degradation in detection accuracy. In particular, in the case where the detector **41** includes an air-suspended microstructure, the microstructure may deform due to the vibration of the gas. This further degrades the detection accuracy.

(32) The resonance frequency f (hereafter also simply referred to as “frequency f ”) of the Helmholtz resonance phenomenon depends on the volume V of the gas detection space **42**, the effective length L of the hole **43**, and the cross-sectional area S of the hole **43**. A frequency band with high unwanted sound intensity due to music or voice in an indoor environment is a band of 100 Hz to 2.0 kHz, and has a peak around 500 Hz in particular. A frequency band with high unwanted sound intensity due to the operation of an air conditioner and the like is a band of 10 Hz to 300 Hz, and has a peak around 100 Hz in particular. As mentioned above, the apparatus **1** provides the acoustic function or the air conditioning function, and emits unwanted sound from the vibration source **20**. In the case where the resonance frequency f in the gas measurement unit **40** coincides with the peak of the frequency band with high unwanted sound intensity, the Helmholtz

resonance phenomenon is likely to be excited. In such a case, the detector **41** is affected and the detection accuracy degrades. In this embodiment, the structure of the gas measurement unit **40** is designed to separate the resonance frequency f from the peak wavelength of unwanted sound intensity so that the Helmholtz resonance phenomenon will not be excited. Moreover, the resonance frequency f is high in the case where the hole **43** is significantly large relative to the volume V of the gas detection space **42**, according to the following Formula (9). In such a case, the detection target gas increased in temperature, pressure, etc. due to the detector **41** and the detection target gas present outside the gas detection space **42** mix quickly through the hole **43**. This makes measurement difficult. Therefore, the resonance frequency f needs to be lower than a certain frequency. That is, the structure of the gas measurement unit **40** is designed so that the resonance frequency f will satisfy the condition “500 Hz < f < 1 MHz”. More preferably, the structure of the gas measurement unit **40** is designed to separate the resonance frequency f from the peak wavelength of unwanted sound intensity including harmonics so that the condition “1.0 kHz < f < 500 kHz” will be satisfied. Further preferably, the structure of the gas measurement unit **40** is designed to separate the resonance frequency f from the frequency band with high unwanted sound intensity so that the condition “2.0 kHz < f < 200 kHz” or “1 Hz < f < 100 Hz” will be satisfied. The frequency f is given by the following Formula (9).

$$(33) \quad f = \frac{c}{2} \sqrt{\frac{S}{VL}}. \quad \text{Formula(9)}$$

(34) FIG. 4 illustrates an example of numerical calculation results of noise intensity brought to the detector **41** due to the Helmholtz resonance phenomenon in the case of changing the resonance frequency f of the gas measurement unit **40**. As a model of the sound emitted from the vibration source **20**, the long-term average speech spectrum for Japanese of male and female talkers under the “loud” condition in FIG. 3 in reference document (Kimio Shiraishi et. al. “Amplification rationale for hearing aids based on characteristics of the Japanese language” May 15, 2021) was used. Assuming that the sound passes over the hole **43** along the gas measurement unit **40**, the air vibration energy generated in the gas measurement unit **40** in the case of changing the shape of the gas measurement unit **40** to change the resonance frequency f was simulated, and the noise energy intensity brought to the detector **41** was calculated. In a cavity with a hole where the Helmholtz resonance phenomenon occurs, Lorentz distribution type excitation behavior is exhibited with respect to the resonance frequency f determined by the shape of the cavity, as in the case of forced vibration caused by application of an external force. Particularly in the case where the energy loss in the cavity is little, the excitation behavior approaches delta-function-like excitation characteristics as a result of picking up specific sound near the resonance frequency f . The model of the sound emitted from the vibration source **20** has a sharp intensity peak at around 500 Hz. The same applies to the noise energy intensity. The noise intensity can be rapidly reduced by separating the resonance frequency f away from 500 Hz.

(35) FIG. 5 illustrates another example of numerical calculation results of noise intensity brought to the detector **41** due to the Helmholtz resonance phenomenon in the case of changing the resonance frequency f of the gas measurement unit **40**. As a model of the sound emitted from the vibration source **20**, measurement data of unwanted sound from the air blower during the heating operation of the Hitachi room air conditioner “RAS-AJ36D(W)” indoor unit was used. As in the example in FIG. 4, assuming that the sound passes over the hole **43** along the gas measurement unit **40**, the air vibration energy generated in the case of changing the shape of the gas measurement unit **40** to change the resonance frequency f was simulated, and the noise energy intensity brought to the detector **41** was calculated. The model of the sound emitted from the vibration source **20** has a sharp intensity peak at around 100 Hz. The same applies to the noise energy intensity. The noise intensity can be rapidly reduced by separating the resonance frequency f away from 100 Hz.

(36) In the case where the hole **43** is provided with the vent pipe **44**, the cross-sectional area S is the average cross-sectional area of the hole **43** and the vent pipe **44**. Moreover, c is the sound

speed. For example, c is 337.7 m/s in the case of air at 10° C., 346.5 m/s in the case of air at 25° C., and 349.4 m/s in the case of air at 30° C., and is approximated by $331.5+0.61t$ [m/s] where t is the Celsius temperature. In the case where the concentration of the detection target gas is low, these values may be appropriately selected and used according to the room temperature. For example in the case where the concentration of the detection target gas is high, the sound speed c is given by the following Formula (10), where ρ is the density of the gas in the gas detection space **42**, γ is the specific heat ratio, and p is the pressure.

$$(37) \quad c = \sqrt{\frac{\gamma p}{\rho}}. \quad \text{Formula(10)}$$

<Effective Length L >

(38) The effective length L of the hole **43** varies depending on the shape of the hole **43** and the length L' that is the total length of the hole **43** and the vent pipe **44**. In the case where the shape of the opening is circular, the effective length L is given by the following Formula (11), where a is the radius.

$$L=L'+1.5a \quad \text{Formula (11).}$$

(39) In the case where the exit of the hole **43** and the vent pipe **44** is flat, the effective length L is given by the following Formula (12).

$$L=L'+1.7a \quad \text{Formula (12).}$$

(40) In the case where the cross-sectional shape of the hole **43** and the vent pipe **44** is irregular, the effective length L is given by the following Formula (13), where $S_{\text{sub.out}}$ is the area of the exit opening of the hole **43** and the vent pipe **44**.

$$L=L'+\sqrt{S_{\text{sub.out}}} \quad \text{Formula (13).}$$

(41) For the effective length L , addition according to the shape of the opening of the hole **43** is made to the length L' obtained by combining the lengths of the hole **43** and the vent pipe **44**. The addition corresponds to open end correction. In addition to the gas vibrated in the vent pipe **44**, the gas around the opening also contributes to vibration. Open end correction is therefore necessary. The vibration of the gas around the opening depends on the area of the exit opening, and the gas up to the height corresponding to the square root of the area $S_{\text{sub.out}}$ contributes to vibration, as indicated in Formula (13).

(42) The Helmholtz resonance phenomenon occurs in the case where the hole **43** is small to some extent. The Helmholtz resonance phenomenon also occurs in the case where the amount of gas in the gas detection space **42** as an air spring is greater than the amount of gas contributing to vibration as a mass. In the apparatus **1** with the gas detection function, the volume V of the gas detection space **42** is large, and the following Formula (14) holds in this embodiment.

$$V>10LS \quad \text{Formula (14).}$$

(43) In order for the detector **41** to detect the detection target gas, the volume V of the gas detection space **42** needs to be sufficiently larger than the volume $V_{\text{sub.sen}}$ of the detector **41**. The following Formula (15) holds in this embodiment, where the volume $V_{\text{sub.sen}}$ of the detector **41** is the outer size of the detector **41** and is the volume of the minimum rectangular parallelepiped including the detector **41**.

$$V>2V_{\text{sub.sen}} \quad \text{Formula (15).}$$

(44) In the case where the ratio of the detector **41** to the gas detection space **42** is high, the ratio of absorption of vibration energy by Helmholtz resonance occurring in the gas measurement unit **40** is high, so that the influence of noise is significant in a small gas sensor. With such design that sets the resonance frequency f outside the unwanted sound band, the detection accuracy can be improved significantly. That is, in this embodiment, the following Formula (16) holds and desirably the following Formula (17) holds for the volume V of the gas detection space **42** relative to the volume $V_{\text{sub.sen}}$ of the detector **41**.

$$V<400V_{\text{sub.sen}} \quad \text{Formula (16).}$$

$$v<100V_{\text{sub.sen}} \quad \text{Formula (17).}$$

(45) Particularly in the case where a thermal light source is used in the detector **41**, for example, intermittent operation by turning the light source on and off at high speed cannot be performed, and accordingly the chopping frequency by chopping control cannot be increased. It is therefore difficult to separate, by frequency, the noise caused by unwanted sound and the output signal modulated by chopping. With such design that sets the resonance frequency f outside the unwanted sound band, the detection accuracy can be improved significantly. That is, in the case where the chopping frequency f_c is less than 500 Hz, such design that sets the resonance frequency f outside the unwanted sound band can improve the detection accuracy significantly. In the case where a thermal light source is not used, too, the detection accuracy can be improved significantly without performing high-speed chopping control that requires a dedicated control IC for which high design technology and cost are needed.

(46) <Arrangement of Vibration Source and Detector in Housing>

(47) As described above, with such design that limits the range of the frequency f , the gas measurement unit **40** achieves higher detection accuracy for the detection target gas. To further enhance the detection accuracy, it is preferable to also take into account the vibration transmission path from the vibration source **20**. For example, the vibration generated by the vibration source **20** is also transmitted to the detector **41** through the gas in the housing **10** and the substrate **30**. It is therefore preferable to adjust the positional relationship between the vibration source **20** in the housing **10** and the detector **41** on the substrate **30** so as to weaken the vibration transmitted to the detector **41**.

(48) First, consider the transmission path through the gas in the housing **10**. It is assumed that the housing **10** behaves as a resonance box. The vibration of the gas in the housing **10** increases due to the resonance mode vibration. If the vibration source **20** is located at a crest-trough position (i.e. a position with large variation) in the resonance mode, large vibration is applied to the detector **41**, resulting in degradation in detection accuracy. To enhance the detection accuracy for the detection target gas, it is preferable to separate the vibration source **20** away from the crest-trough position in the resonance mode of the housing **10** and place the vibration source **20** at a node position. Since the shape is definite in the low-order mode of the resonance mode, the node position can be determined in terms of design. Specifically, in the case where the housing **10** has a rectangular parallelepiped shape, a cylindrical shape, or a spherical shape, the plane of symmetry, the axis of symmetry, or the center point is a node in the resonance mode of sound pressure. For example, the center between a pair of facing surfaces is a node. By placing the vibration source **20** at such a position, the amount of vibration transmitted from the vibration source **20** to the detector **41** is reduced, with it being possible to improve the detection accuracy of the apparatus **1**.

(49) The crest-to-crest length in the resonance mode is determined by the size of the housing **10**. Let $V_{sub.hus}$ be the volume of the housing **10**, and typical length $L_{sub.hus}$ be the cube root of the volume $V_{sub.hus}$ or the distance from the inner wall surface of the housing **10** to the node. Then, the detector **41** is placed at a position of $L_{sub.hus}/10$ from the node position in the length in a certain direction. The typical length $L_{sub.hus}$ varies according to the shape of the housing **10**, and may be the cube root in the case where the side lengths of the housing **10** are equal in three directions, and the distance from the inner wall surface to the node in the case where the housing **10** has any other shape.

(50) The edge parts of the housing **10** are each a crest position in the resonance mode. By placing the vibration source **20** at a position other than the edge parts of the housing **10**, the amount of vibration transmitted from the vibration source **20** to the detector **41** is reduced, with it being possible to improve the detection accuracy of the apparatus **1**. For example, the detector **41** is placed at a position inward from an edge of the housing **10** by more than $L_{sub.hus}/5$.

(51) The housing **10** may have a shape more complex than a rectangular parallelepiped, a cylinder, or a sphere. Since the behavior in the low-order resonance mode is qualitatively the same, however, it suffices to approximate the housing **10** by a rectangular parallelepiped, a cylinder, or a sphere so

as to minimize the volume error, and place the vibration source **20** or the detector **41** not at a crest position but at a node position in the defined shape. For example, the vibration source **20** or the detector **41** may be located at the plane of symmetry (node position) of a rectangular parallelepiped approximately defined for the housing **10**. For example, the vibration source **20** or the detector **41** may be located at the axis of symmetry (node position) of a cylinder approximately defined for the housing **10**. For example, the vibration source **20** or the detector **41** may be located at the center point (node position) of a sphere approximately defined for the housing **10**.

(52) <Arrangement of Vibration Source and Detector on Substrate>

(53) In the case where the periphery of the substrate **30** is fixed, the substrate **30** behaves as a resonance plate whose periphery does not vibrate. If the vibration source **20** is located at a crest-trough position in the resonance mode, energy is efficiently transferred for the resonance mode vibration of the vibration of the substrate **30**. In such a case, the vibration of the substrate **30** increases, and the vibration is applied to the detector **41** through the substrate **30**. The vibration is detected by the detector **41** as an electric signal, resulting in degradation in detection accuracy. Moreover, if the detector **41** is located at a crest-trough position in the resonance mode, large vibration is applied to the detector **41**. The vibration is detected by the detector **41** as an electric signal, resulting in degradation in detection accuracy.

(54) From the resonance mode analysis of a rectangular plate or a circular plate, each end or the axis of symmetry of the substrate **30** is a node. By placing the vibration source **20** or the detector **41** at such a node position, the amount of vibration transmitted from the vibration source **20** to the detector **41** is reduced, with it being possible to improve the detection accuracy of the apparatus **1**.

(55) The crest-to-crest length in the resonance mode is determined by the size of the substrate **30**. Let S_{sub} be the area of the substrate **30**, and typical length L_{sub} be the square root of the area S_{sub} . Then, the vibration source **20** or the detector **41** is placed at a position of $L_{sub}/10$ from the node position.

(56) The center part of the substrate **30** is a crest position in the resonance mode. By placing the vibration source **20** at a position other than the center part of the substrate **30**, the amount of vibration transmitted from the vibration source **20** to the detector **41** is reduced, with it being possible to improve the detection accuracy of the apparatus **1**. For example, the vibration source **20** or the detector **41** is placed at a position outward from the center part of the substrate **30** by more than $L_{sub}/10$.

(57) The substrate **30** may have a shape more complex than a rectangular or circular plate. Since the behavior in the low-order resonance mode is qualitatively the same, however, it suffices to approximate the substrate **30** by a rectangular or circular plate so as to minimize the volume error, and place the vibration source **20** or the detector **41** not at a crest position but at a node position in the defined shape. For example, the vibration source **20** or the detector **41** may be located at an end of a rectangle approximately defined for the substrate **30**.

(58) <Size of Housing>

(59) The housing **10** has a resonance mode of the lowest order of frequency f_{hus} approximately represented by the following Formula (18), from resonance mode analysis in the case of a rectangular parallelepiped. By setting the frequency f_{sub} outside the frequency band with high unwanted sound intensity or the frequency band with high unwanted sound intensity due to driving of an air conditioner, the vibration applied to the detector **41** can be reduced to thus improve the detection accuracy of the apparatus **1** for the detection target gas.

(60) $f_{hus} = \frac{c}{L_{hus}}$. Formula(18)

(61) f_{sub} is preferably 500 Hz or more. f_{sub} is more preferably 1.0 kHz or more. f_{sub} is further preferably 2.0 kHz or more. f_{sub} is preferably 100 Hz or less.

(62) <Size of Substrate>

(63) The substrate **30** has a resonance mode of the lowest order of frequency f_{sub}

approximately represented by the following Formula (19), from resonance mode analysis in the case of a rectangular plate. By setting the frequency f_{sub} outside the frequency band with high unwanted sound intensity or the frequency band with high unwanted sound intensity due to driving of an air conditioner, the vibration applied to the detector **41** can be reduced to thus improve the detection accuracy of the apparatus **1** for the detection target gas.

$$(64) \quad f_{\text{sub}} = \frac{h}{2L_{\text{sub}}^2} \sqrt{\frac{Eg}{3(1-\sigma^2)}} \quad \text{Formula(19)}$$

(65) Here, E is the modulus of longitudinal elasticity, g is the gravitational acceleration and is given by 9.80665 m/s², σ is Poisson's ratio, h is the thickness of the substrate **30**, and γ is a dimensionless constant given by 3.65. f_{sub} is preferably 500 Hz or more. f_{sub} is more preferably 1.0 kHz or more. f_{sub} is further preferably 2.0 kHz or more. f_{sub} is preferably 100 Hz or less.

(66) <Dust Filter>

(67) The hole **43** may be provided with the dust filter **45**. The dust filter **45** reflects or attenuates the vibration (sound) of gas in the gas detection space **42**, thereby suppressing the vibration applied to the detector **41**. The detection accuracy of the apparatus **1** can thus be improved. When the acoustic characteristic impedance of the dust filter **45** is higher, the reflection or attenuation performance of the dust filter **45** is higher. The acoustic characteristic impedance is preferably 4000 kg/m²s or more, which is sufficiently larger than the impedance of air, i.e. approximately 10 times or more the impedance of air. The acoustic characteristic impedance is more preferably 40000 kg/m²s or more. The acoustic characteristic impedance is given by the product of the medium density ρ_m [kg/m³] and the sound speed C_m in the medium.

(68) As described above, the apparatus with the gas detection function according to this embodiment having the foregoing structure can detect the detection target gas with high accuracy by suppressing vibration that affects the gas measurement unit **40**.

(69) While the embodiments of the present disclosure have been described by way of the drawings and examples, various changes and modifications may be easily made by those of ordinary skill in the art based on the present disclosure. Such changes and modifications are therefore included in the scope of the present disclosure. For example, the functions included in the components, etc. may be rearranged without logical inconsistency, and a plurality of components, etc. may be combined into one component, etc. and a component, etc. may be divided into a plurality of components, etc.

Claims

1. An apparatus with a gas detection function, comprising: a housing; a vibration source; and a gas measurement unit located inside the housing and demarcated by a partition, wherein the vibration source is located outside the gas measurement unit, the gas measurement unit includes a detector located on a substrate, and a gas detection space provided with a hole through which a gas passes, and a frequency f expressed by the following Formula (1): $f = \frac{c}{2} \sqrt{\frac{S}{VL}}$ Formula(1) is 500 Hz or more, where V is a volume of the gas detection space, S is a cross-sectional area of the hole, L is an effective length of the hole, and c is a sound speed.

2. The apparatus with a gas detection function according to claim 1, wherein the hole is provided with a vent pipe, and the effective length is given by the following Formula (2):

$L = L' + 1.5a$ Formula (2), where L' is a total length of the hole and the vent pipe, and a is a radius of the hole.

3. The apparatus with a gas detection function according to claim 1, wherein the hole is provided with a vent pipe, an exit of the hole and the vent pipe is flat, and the effective length is given by the following Formula (3):

$L=L'+1.7a$ Formula (3), where L' is a total length of the hole and the vent pipe, and a is a radius of the hole.

4. The apparatus with a gas detection function according to claim 1, wherein the hole is provided with a vent pipe, a cross-sectional shape of the hole and the vent pipe is irregular, and the effective length is given by the following Formula (4):

$L=L'+\sqrt{S_{\text{sub.out}}}$ Formula (4), where L' is a total length of the hole and the vent pipe, and $S_{\text{sub.out}}$ is an area of an exit opening of the hole and the vent pipe.

5. The apparatus with a gas detection function according to claim 1, wherein the frequency f is 1.0 kHz or more.

6. The apparatus with a gas detection function according to claim 1, wherein the frequency f is 2.0 kHz or more.

7. An apparatus with a gas detection function, comprising: a housing; a vibration source; and a gas measurement unit located inside the housing, wherein the gas measurement unit includes a detector located on a substrate, and a gas detection space provided with a hole through which a gas passes, the hole is provided with a vent pipe, a cross-sectional shape of the hole and the vent pipe is irregular, an effective length L of the hole is given by the following Formula (5):

$L=L'+\sqrt{S_{\text{sub.out}}}$ Formula (5), where L' is a total length of the hole and the vent pipe, and $S_{\text{sub.out}}$ is an area of an exit opening of the hole and the vent pipe, and a

frequency f expressed by the following Formula (6): $f = \frac{c}{2} \sqrt{\frac{S}{VL}}$ Formula(6) is 100 Hz or less, where V is a volume of the gas detection space, S is a cross-sectional area of the hole, and c is a sound speed.

8. The apparatus with a gas detection function according to claim 1, wherein the vibration source or the detector is located at a plane of symmetry of a rectangular parallelepiped approximately defined for the housing.

9. The apparatus with a gas detection function according to claim 1, wherein the vibration source or the detector is located at an axis of symmetry of a cylinder approximately defined for the housing.

10. The apparatus with a gas detection function according to claim 1, wherein the vibration source or the detector is located at a center point of a sphere approximately defined for the housing.

11. The apparatus with a gas detection function according to claim 1, wherein the vibration source or the detector is located at an end of a rectangle approximately defined for the substrate.

12. The apparatus with a gas detection function according to claim 1, wherein a frequency $f_{\text{sub.hus}}$ expressed by the following Formula (7): $f_{\text{hus}} = \frac{c}{L_{\text{hus}}}$ Formula(7) is 500 Hz or more, where $L_{\text{sub.hus}}$ is a typical length of the housing.

13. The apparatus with a gas detection function according to claim 12, wherein the frequency $f_{\text{sub.hus}}$ is 1.0 kHz or more.

14. The apparatus with a gas detection function according to claim 12, wherein the frequency $f_{\text{sub.hus}}$ is 2.0 kHz or more.

15. The apparatus with a gas detection function according to claim 1, wherein a frequency $f_{\text{sub.sub}}$ expressed by the following Formula (8): $f_{\text{sub}} = \frac{h}{2L_{\text{sub}}^2} \sqrt{\frac{E_{\text{g}}}{3(1-\sigma^2)}}$ Formula(8) is 500 Hz or more,

where $S_{\text{sub.sub}}$ is an area of the substrate, E is a modulus of longitudinal elasticity of the substrate, h is a thickness of the substrate, σ is Poisson's ratio, γ is a dimensionless constant of 3.65, and a typical length $L_{\text{sub.sub}}$ is a square root of the area of the substrate.

16. The apparatus with a gas detection function according to claim 15, wherein the frequency $f_{\text{sub.sub}}$ is 1.0 kHz or more.

17. The apparatus with a gas detection function according to claim 15, wherein the frequency $f_{\text{sub.sub}}$ is 2.0 kHz or more.

18. The apparatus with a gas detection function according to claim 1, wherein the hole is provided with a dust filter, and an acoustic characteristic impedance of the dust filter is 4000 kg/m²s or

more.

19. The apparatus with a gas detection function according to claim 1, wherein the detector includes an air-suspended microstructure.

20. The apparatus with a gas detection function according to claim 1, wherein the vibration source is located inside the housing.
